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1 Objectives

The behavior of beams under different loading conditions is a fundamental aspect of structural engineering. Understanding how beams deflect when subjected to varying loads is crucial in designing safe and efficient structures. In this experiment, the focus is on two specific beam configurations: the cantilever and the simply supported beam. The deflection of these beams will be investigated both theoretically and experimentally.

The objectives for this experiment are outlined as follows:

1. **Measurement of Deflection:** Experimentally acquire accurate values of deflection for both a cantilever and a simply supported beam under varying loads to understand their structural behaviors.

2. **Comparison of Experimental and Theoretical Results:** Analyze and compare the obtained experimental deflection values with the theoretically predicted values for both the cantilever and the simply supported beam to assess the accuracy and reliability of theoretical models in predicting structural deflections.

These objectives aim to provide a comprehensive understanding of how different types of beam structures behave under applied loads and to evaluate the alignment between theoretical predictions and actual experimental observations in real-world scenarios.

2 Procedure

This experiment consists of two parts: the cantilever beam testing and the simply supported beam testing.

2.1 Cantilever Beam Testing

1. **Step 1:** Check the initial readings of the electrical potentiometers to ensure they are zero.

2. **Step 2:** Apply loads in increments of 2 lb (8.9 N) at the mid-span ($P1$) up to a total load of 10 lb. Record the vertical deflections at both the mid-span ($P1$) and at the end of the cantilever beam ($P2$) for each load.

3. **Step 3:** Remove the applied load.

4. **Step 4:** Apply the same set of loads to the end of the beam ($P2$) and measure the deflections at the mid-span ($P1$) and at the end of the cantilever beam ($P2$).

2.2 Simply Supported Beam Testing

1. **Step 1:** Ensure that the initial readings of the electrical potentiometers are zero.

2. **Step 2:** Apply loads in increments of 10 lb (44.482 N) at the mid-span of the beam ($P3$) up to a total load of 40 lb (177.928 N). Record the vertical deflections at both the mid-span ($P3$) and at the quarter-span of the beam ($P4$) for each load.

3. **Step 3:** Remove the applied load.

4. **Step 4:** Apply the same set of loads to the quarter-span ($P4$) and measure the deflections at the mid-span ($P3$) and at the quarter-span of the beam ($P4$).

Throughout the experiment, ensure accurate recording of all measurements, maintain the safety protocols, and perform the tests as per the specified loading conditions for each beam configuration.

2.3 Free-Body Diagrams

This section contains the free-body diagrams for the each beam and each loading condition.

2.3.1 Cantilever Beam Load Applied at the Mid-Span

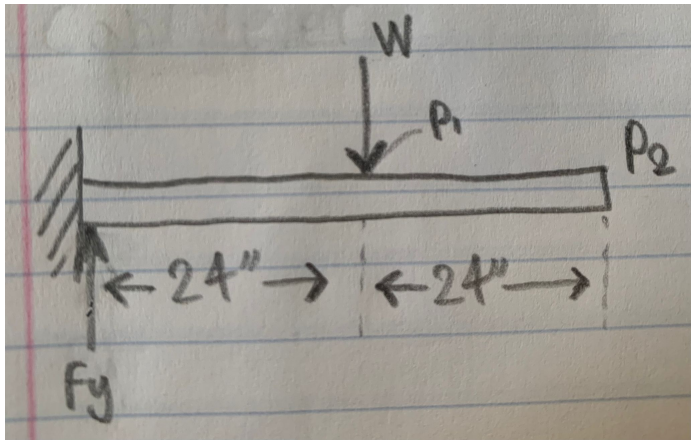

Figure 1: Free-body diagram of the cantilever beam with the load applied at the mid-span.

2.3.2 Cantilever Beam Load Applied at the End

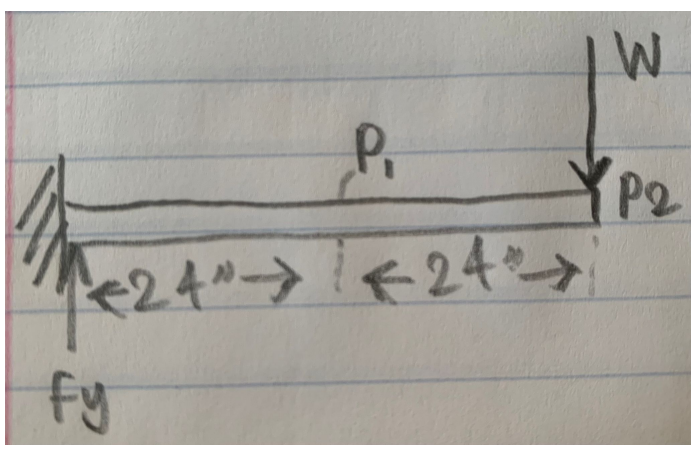

Figure 2: Free-body diagram of the cantilever beam with the load applied at the end.

2.3.3 Simply Supported Beam Load Applied at the Mid-Span

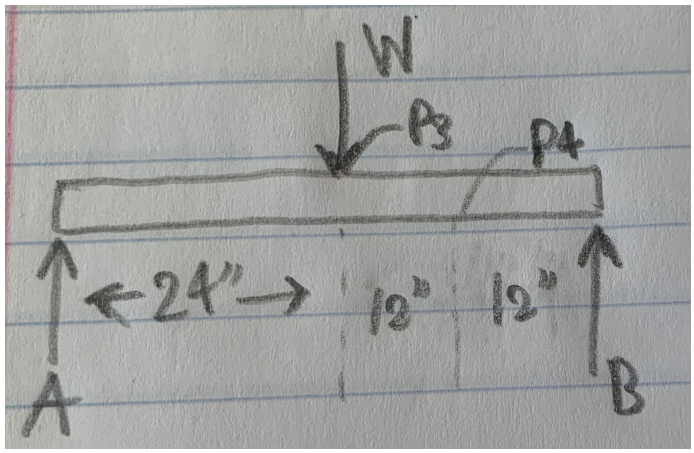

Figure 3: Free-body diagram of the simply supported beam with the load applied at the mid-span.

2.3.4 Simply Supported Beam Load Applied at the Quarter-Span

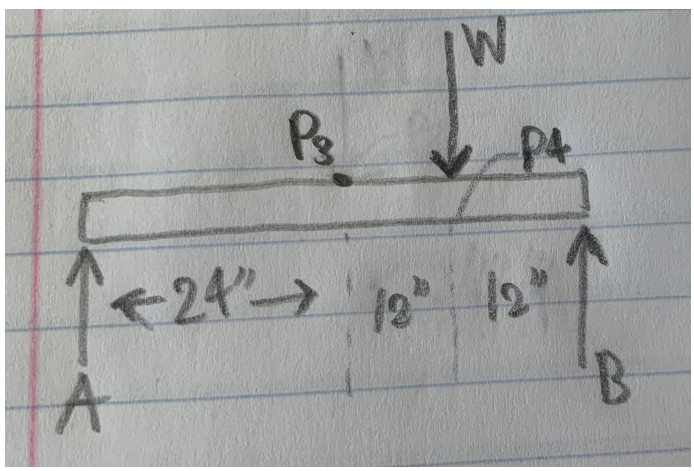

Figure 4: Free-body diagram of the simply supported beam with the load applied at the quarter-span.

3 Results

This section contains theoretical derivation of the deflection equations for each beam and each loading condition, the experimental results, and the plots and summary tables of the theoretical and experimental results in the comparison section.

3.1 Cantilever Beam Load Applied at the Mid-Span

3.1.1 Theoretical Results

3.1.2 Experimental Results

3.1.3 Comparison of Theoretical and Experimental Results

3.2 Cantilever Beam Load Applied at the End

3.2.1 Theoretical Results

3.2.2 Experimental Results

3.2.3 Comparison of Theoretical and Experimental Results

3.3 Simply Supported Beam Load Applied at the Mid-Span

3.3.1 Theoretical Results

3.3.2 Experimental Results

3.3.3 Comparison of Theoretical and Experimental Results

3.4 Simply Supported Beam Load Applied at the Quarter-Span

3.4.1 Theoretical Results

3.4.2 Experimental Results

3.4.3 Comparison of Theoretical and Experimental Results

4 Discussion

4.1 Analysis of Results

4.2 Sources of Error

5 Conclusions

5.1 Summary

5.2 Maxwell's Law of Reciprocal Deflections